Compact size 10/100M Industrial Media Converter

LT-IMC30F-E Series



Features:

- 1*10/100Base-TX RJ45 port, 1*100Base-FX SFP(/SC/ST/FC optional) fiber port
- DC 12-58V input, redundant power supply with polarity reverse protection
- Powerful Dip switch function:
  1. LFP; 2. Flow control; 3. Broadcast storm restrain; 4.10Kbytes jumbo frame
- Support watchdog, it can reboot automatically if the device is frozen
- Support 1M large buffer
- Support 4KV surge protection and ESD: Air-15kV, Contact-8kV Protection
- IP40 fan-less and Din-rail hardware design with compact size (30*85*85mm)
- Operation temperature: -40 ℃ ~+75℃

Overview

The SmartByte LT-IMC30F-E series are unmanaged industrial grade media converter with 1-port 10/100Base-TX RJ45 and 1-port 100Base-FX fiber optical interface. They are featuring with LFP, flow control, broadcast storm restrain and 10Kbytes jumbo frame function, which all can be configured by the Dip switch on the top panel.

LT-IMC30F-E series are also high cost-effective easy-to-use device, which provide essential industrial Ethernet networking function, such as wide range power input 12-58VDC, redundant power design with polarity reverse protection, robust IP40 fan-less compact size housing with Din-rail installation, wide operation temperature from -40 ℃ to 75 ℃ as well as high-level EMI/EMC capability and so on. They are the best choice for heavy industrial factory, transportation, oil & gas, chemical, IP Surveillance and processing automation area where environmental conditions exceed commercial product specifications

Technical specification

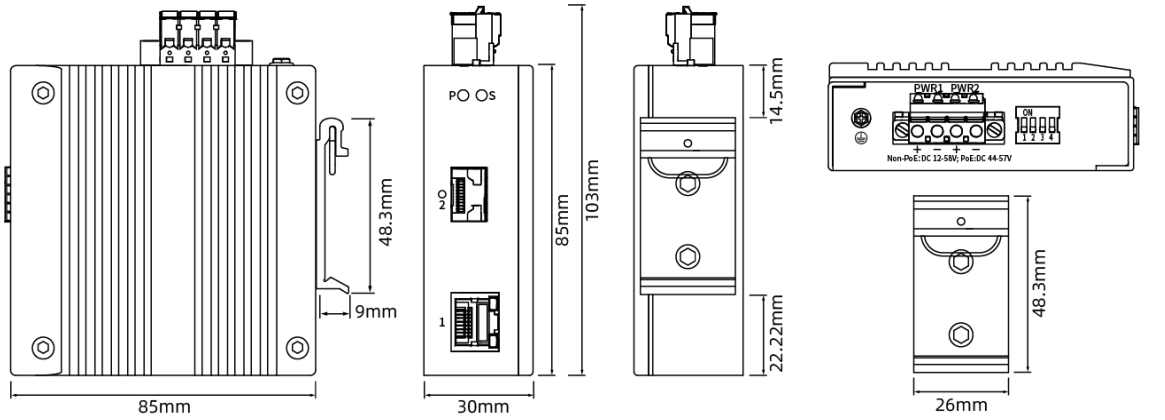
<table>
<tr><td>Model No.</td><td colspan="3">LT-IMC30F-E</td></tr>
<tr><td rowspan="2">Interface</td><td>Fiber port</td><td colspan="2">Copper RJ45 port</td></tr>
<tr><td>1</td><td colspan="2">1</td></tr>
<tr><td>Ethernet</td><td colspan="3">1*10/100Base-TX RJ45
1*100Base-FX SFP (SC/ST/FC optional)</td></tr>
<tr><td>Standard</td><td colspan="3">IEEE 802.3 10Base-T Ethernet
IEEE 802.3u Fast Ethernet
IEEE 802.3x Full-Duplex Flow Control</td></tr>
<tr><td>Dip Switch</td><td colspan="3">1. LFP
2. Flow control
3. Broadcast storm restrain
4. 10Kbytes jumbo frame</td></tr>
<tr><td rowspan="8">LED Indicators</td><td rowspan="2">P (Power indicator) Green</td><td colspan="2">Off: the device is power off or failed</td></tr>
<tr><td colspan="2">On: the device power on is normal</td></tr>
<tr><td rowspan="2">S (System indicator) Red</td><td colspan="2">Off: the chip is normal</td></tr>
<tr><td colspan="2">On: the chip read/write is unnormal</td></tr>
<tr><td rowspan="4">1 Copper port)</td><td>Green indicators</td><td>Yellow indicators</td></tr>
<tr><td>Off: ports link down</td><td>Off: port speed is 10M</td></tr>
<tr><td>On: ports link up</td><td rowspan="2">On: port speed is 100M</td></tr>
<tr><td>Blinking: data on TX/RX</td></tr>
<tr><td>2 (Fiber port) Green</td><td colspan="2">Off: ports link down</td></tr>
</table>

		On: ports link up
		Blinking: data on TX/RX
Power parameters		
Input voltage	12-58VDC, redundant power input	
Input current	0.2A Max	
Total power consumption	Full loading ≤2W	
Connector	Removable 4-pin terminal block	
Reverse polarity protection	Support	
Switching features		
Switching capacity	0.4G	
Packet forwarding rate	0.59Mpps	
MAC address table	8K	
VLAN	Support the transmission of the packets with VLAN tags ranging from 1 to 4096	
Buffer	1M	
Forwarding delay	<10us	
Jumbo Frame	Support 10Kbytes	
MDI/MDIX	Support	
Watchdog	Support	
Mechanical structure		
Case protection	IP40	
Installation method	Din-rail	
Dimension(W*D*H)mm	30*85*85mm	
Weight	0.2 kg	
Operating environment		
Operating temperature	-40℃~+75℃	
Storage/transportation temperature	-40℃~+85℃	
Relative humidity	5%~95%（non-condensing）	
Industrial Standard	Surge protection of power: IEC 61000-4-5 Level 3 (4KV/2KV) （8/20us）	
	Surge protection of Ethernet ports: IEC 61000-4-5 Level 3 (4KV/2KV) （10/700us）	

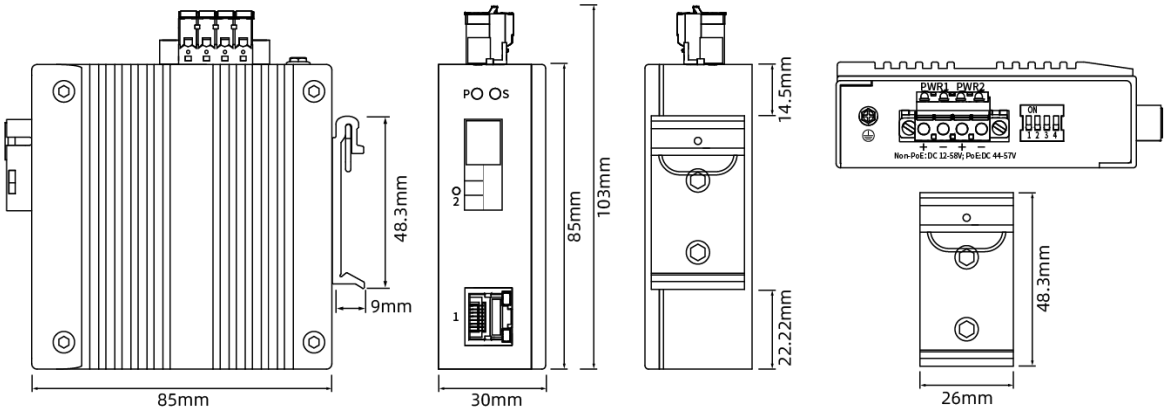
	DIP: IEC 61000-4-11　Level 3（10V）
	ESD: IEC 61000-4-2　Level 4（8K/15K）
	Shock：IEC 60068-2-27
	Free fall：IEC 60068-2-32
	Vibration：IEC 60068-2-6
Certification	CCC/CE/FCC/RoHS/UKCA
MTBF	> 200,000 hours
Warranty	5 years

Structure diagram

SFP model



SC/ST/FC model



Order information

Model	Description
LT-IMC30F-E	10/100Base-TX to 100Base-FX unmanaged compact size industrial media converter with 1*10/100Base-TX RJ45 port and 1*100Base-FX SFP slot, DC12-58V input, redundant dual power supply, Din-rail installation. Operation temperature: -40℃～+75℃
LT-IMC30F-SC/ST/FC-E	10/100Base-TX to 100Base-FX unmanaged compact size industrial media converter with 1*10/100Base-TX RJ45 port and 1*100Base-FX SC port (ST and FC optional), DC12-58V input, redundant dual power supply, Din-rail installation; Multi-mode/Single mode/ dual fiber/single fiber, 2KM -120KM optional; Operation temperature: -40℃～+75℃